Unit 15, Lesson 1: Exploring Sample Space

Benchmark

MA.7.DP.2.1 Determine the sample space for a simple experiment.

Learning Targets

- ☐ I can determine all possible outcomes of a given scenario.
- ☐ I can determine whether or not a simple experiment is fair.

15.1.1 Warm-Up: Carolyn Beatrice Parker

Carolyn Beatrice Parker, born in Gainesville, FL in 1917, was the first African American woman known to receive a graduate degree in physics and subsequently work on the top-secret government project known as the Manhattan Project. Before earning her master's degree in physics, she earned a master's degree in mathematics and was also a teacher in Florida.

In 2020, an elementary school in Gainesville, FL was re-named in her honor. At this time, 11 of the 21 elementary schools in the Alachua County Public School system were named after people.

1. Without performing any calculations, select the statement that best describes the percentage of the elementary schools in Alachua County that were named after people as of 2020.

   - ○ significantly less than 50%
   - ○ slightly less than 50%
   - ○ exactly 50%
   - ○ slightly more than 50%
   - ○ significantly more than 50%

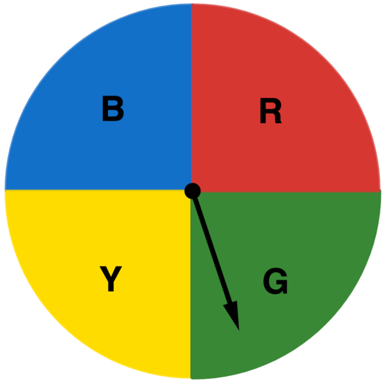
2. Explain your reasoning.

15.1.2 Guided Instruction: Determining Outcomes

An ***outcome*** is a possible result of an experiment.

1. A spinner divided into four equal areas is shown.

   a. How many outcomes are there from spinning the spinner?



In a probability model for a random process, the ***sample space*** is a list of the individual outcomes that are being considered.

   b. With your partner, discuss the sample space for the spinner. Write a list of the possible outcomes.

An ***event*** is a set of possible outcomes resulting from an experiment. In general, an event is any subset of a sample space.

   c. How many outcomes are included in the event that the spinner lands on blue?

   d. Primary colors include yellow, blue, and red. How many outcomes are included in the event that the spinner lands on a primary color?

2. Doug's parents created a set of cards with chores on them. Each Saturday, Doug chooses a card to determine his chore.

Mow the yard	Dust and vacuum the house	Clean the bathrooms	Clean your bedroom
Clean your bedroom	Mow the yard	Weed the garden	Mow the yard
Clean the bathrooms	Parents' choice	Dust and vacuum the house	Free day

a. What are the possible outcomes for the chore cards?

b. Which outcomes have more than one card?

c. On a particularly hot Saturday, Doug is hoping he doesn't select an outdoor chore. Which outcomes could be included in the event that he chooses an outdoor chore?

d. How many outcomes could be included in the event that Doug chooses an indoor chore?

3. Use the word bank to complete the summary of key concepts below.

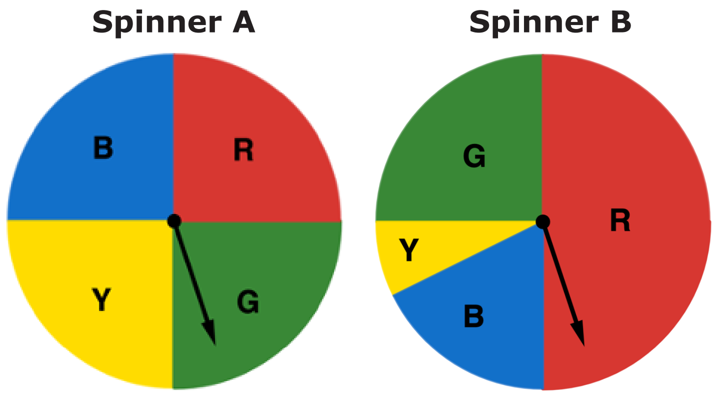
event	outcomes	sample space	set

To determine the ____________________ of a simple experiment, find all the ____________________ that are possible and write them as a ____________________.

An ____________________ is a subset of the sample space, such as rolling an even number on a 6-sided die.

15.1.3 Exploration: Digging Deeper Into Outcomes

Two spinners are shown.



1. Work with your partner to compare and contrast the features of the two spinners.

Similarities	Differences

2. One of the spinners is considered a fair spinner. Complete the sentence by selecting the fair spinner and writing an explanation.

- ○ Spinner A
- ○ Spinner B

is the fair spinner because ______________

____.

15.1.4 Activity: Gallery Walk

1. Work with your partner to determine the sample space for each game and whether the game is fair. Fill in the chart with your answers.

Game Name	Sample Space	Is it fair?
Skee-Ball		
Card Draw		
Prize Wheel		
Rubber Ducks		

2. Which game has the largest sample space?

3. Discuss with your partner which game's sample space was the most difficult to determine.

4. Carnivals include many games similar to these. What do you notice about the fairness of such games?

5. Find another pair of partners to form a group of four. Compare your responses on whether each game is fair and discuss your reasoning. Record one similarity and one difference you notice in your reasoning.

Similarity	**Difference**

6. Future lessons in this unit will explore probabilities with fair coins, fair dice, and fair spinners. Describe what you think it means for a coin, a die, or a spinner to be fair.

15.1.5 Wrap Up: Reflection

Complete each statement based on your understanding at this point.

1. I am still unsure about ______________________________

____

______________________________.

2. One question I have that will help me better understand

is ______________________________

____

______________________________?